AD Patel Institute of Engineering and Technology

Department of Electronics and Communication

Vision

Nurturing excellence in Electronics and Communication Engineering through innovative, rational and ethical thinking.

Mission

- To impart quality education through fundamental theory and praxis in Electronics and Communication Engineering.
- To provide conducive learning environment for holistic development.

Program Educational Outcome [PEOs]

- To provide excellent academic environment to produce graduates who will become successful performers in Electronics and Communication Engineering and other diverse fields.
- To groom students for higher education and research in the core area relevant to their careers with strong mathematical and engineering background.
- To train the students professionally to work as a team leader or as a team member to comprehend, analyze, design and create innovative solutions of real time problems.
- To inculcate positive attitude, effective communication skills, responsibility, accountability and aptitude to work in multi-cultural domain in students.

Program Specific Outcomes [PSOs]

- Gain the knowledge of various analog and digital electronics and communication systems.
- Use the software and hardware based approach in design, analysis and implementation of circuits and systems.
- Create, think and develop an electronics and communication based systems for different applications

AD Patel Institute of Engineering and Technology

Department of Electronics and Communication

A.Y. 2026-27 (ODD), Semester: 3
Subject Name: Signals and Systems
Index

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ENROLMENT NO: 12502060601022

BRA NCH: Electronics and communication Engineering

Sr. No	Name of the Experiment	Date of Experiment	Marks	Signature	Date of Submission
1	Introduction of MATLAB and various commands for Signal and System perspective.				
2	Generate and plot basic signals – Impulse, Unit Step, Ramp, Signum using MATLAB/Scilab. Further generate and plot various complex signals using time shifting, scaling and time reversal properties.				
3	Generate the following continuous time domain signal and plot using MATLAB/Scilab (i) Sine/Cosine signal (ii) Sinc function (iii) Triangle waveform (iv) Rectangular Pulse (v) Sawtooth Waveform (vi) Exponential function				
4	To study and verify the following properties of signal/system using MATLAB/Scilab (i) Linearity (ii) Stability (iii) Time invariant. Find the Energy and Power of the given signal. Separate Even and odd components of given signal and plot respectively.				

5	To Study convolution of following different sequences using MATLAB/Scilab (without using conv command) and plot the resultant signal. (i) Convolution between two rectangular signals. (ii) Convolution between rectangular and Triangular signals. (iii) Convolution between two Triangular signals.				
6	To Study the Fourier series of Square, Sawtooth and Triangular waves using MATLAB/Scilab.				
7	To Study Fourier Transform of a signal using FFT and IFFT command and plot magnitude and phase spectra of a signal using MATLA/Scilab.				
8	To study Convolution Property of Fourier transform for signal using MATLAB / Scilab.				
9	To study Correlation Auto correlation of signal using MATLAB/Scilab				
10	To Study Z Transform and plot region of Convergence using MATLAB/Scilab.				

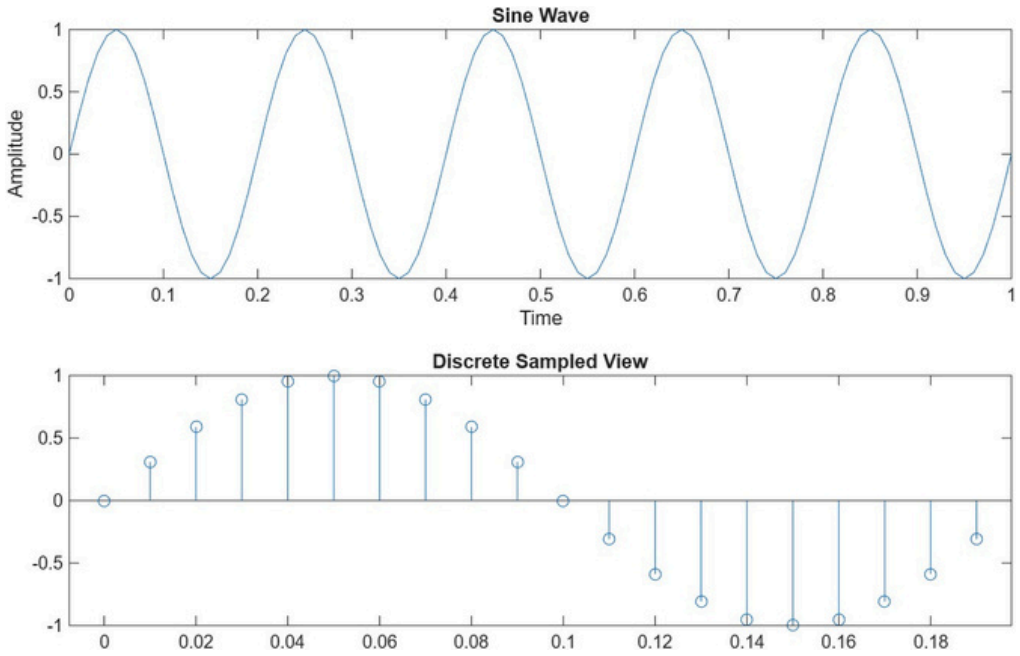
EXPERIMENT NO.1

Aim: To study the MATLAB environment and familiarize with basic MATLAB commands used for signal generation, manipulation, visualization, and system analysis.

Code:

```
clc;
clear;
a = 5;
b = 10;
c = a + b;
t = 0:0.01:1;
f = 5;
x = sin(2 * pi * f * t);
t = 0:0.01:1;
x1 = sin(2 * pi * 5 * t);
x_scaled = 2 * x1;
x_shifted = sin(2 * pi * 5 * (t - 0.1));
x_sum = x1 + x_scaled;
t = 0:0.01:1;
x = sin(2 * pi * 5 * t);
figure;
subplot(2, 1, 1);
plot(t, x);
title('Sine Wave');
xlabel('Time');
ylabel('Amplitude');
subplot(2, 1, 2);
stem(t(1:20), x(1:20));
title('Discrete Sampled View');
```

Output:



EXPERIMENT NO.2

Aim: Generate and plot basic signals – Impulse, Unit Step, Ramp, Exponential , Sine using MATLAB/Scilab. Further generate and plot various complex signals using time shifting, scaling and time reversal properties.

Code:

```
n = -10:1:10;
x1 = 1*(n>=-3 & n<=3);
figure;
subplot(2,3,1); stem(n,x1);
xlabel("sample");
ylabel("amplitude");
title("Rectangular pulse");
x2 = cos(0.3*pi*n);
subplot(2,3,2); stem(n,x2);
xlabel("sample");
ylabel("amplitude");
title("Cosine wave");
x3 = 0.5.^(n).*(n>=0);
subplot(2,3,3); stem(n,x3);
xlabel("sample");
ylabel("amplitude");
title("Exponential decay");
x4 = sin(0.2*pi*n);
subplot(2,3,4); stem(n,x4);
xlabel("sample");
ylabel("amplitude");
```

```
title("Sine wave");
x5 = (n==-4);
subplot(2,3,5);
stem(n,x5);
xlabel("sample");
ylabel("amplitude");
title("Shifted impulse");
x6 = sign(n);
subplot(2,3,6);
stem(n,x6);
xlabel("sample");
ylabel("amplitude");
title("Signum function");

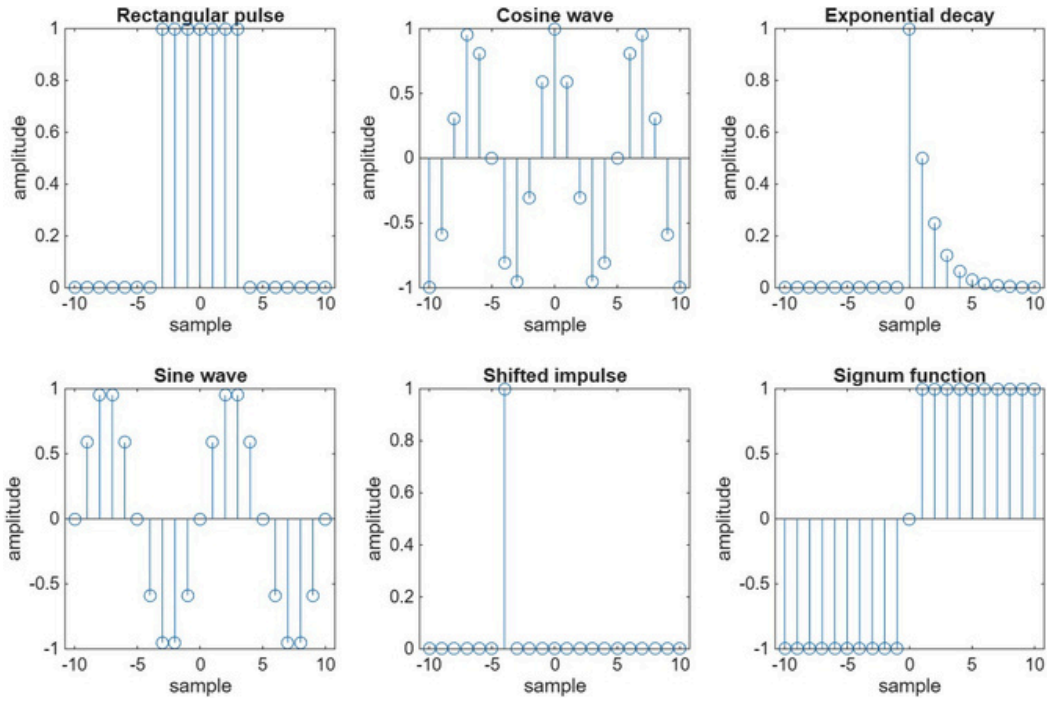

n= -10:1:10;
x= cos(0.2*pi*n) .* (n>=0);
figure;
subplot(2,2,1);
stem(n,x);
xlabel("sample");
ylabel("amplitude");
title("Base signal");

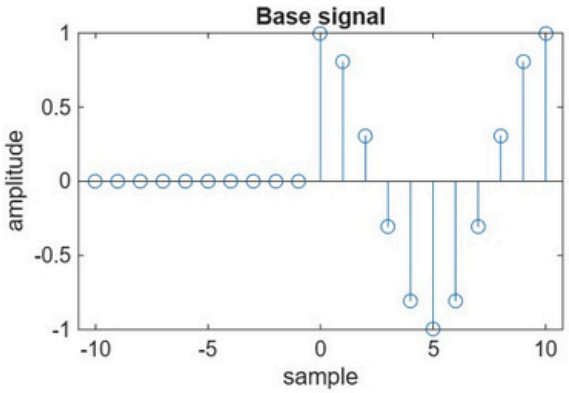

x_shifted = cos(0.2*pi*(n-3)) .* ((n-3)>=0);
subplot(2,2,2);
stem(n,x_shifted);
xlabel("sample");
ylabel("amplitude");
title("Delayed by 3");
```

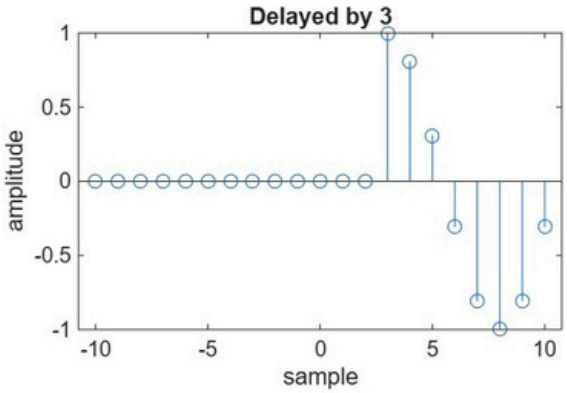
```
x_scaled = cos(0.2*pi*(2*n)) .* ((2*n)>=0);
subplot(2,2,3);
stem(n,x_scaled);
xlabel("sample");
ylabel("amplitude");
title("Compressed signal");

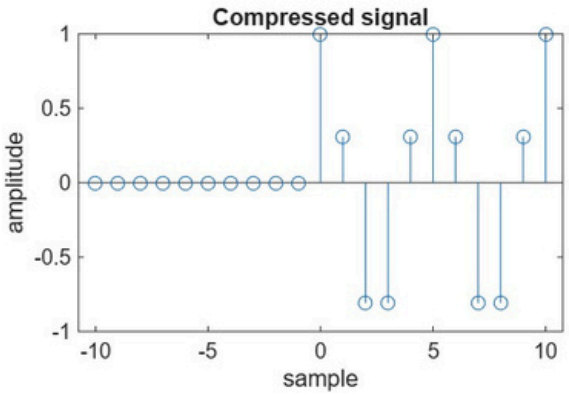
x_reversed = cos(0.2*pi*(-n)) .* ((-n)>=0);
subplot(2,2,4);
stem(n,x_reversed);
xlabel("sample");
ylabel("amplitude");
title("Time-folded signal");
```

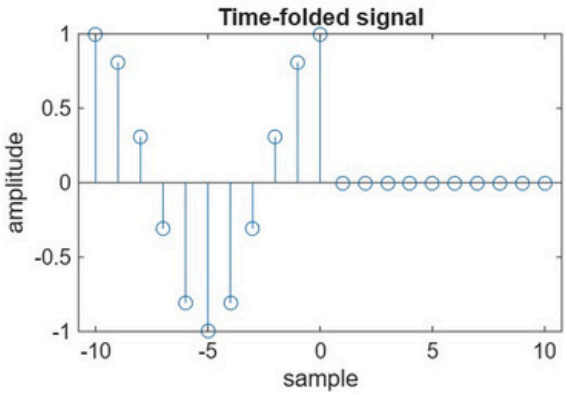
Output:




Base signal
amplitude
sample
1
0.5
0
-0.5
-1
-10
-5
0
5
10


Delayed by 3
amplitude
sample
1
0.5
0
-0.5
-1
-10
-5
0
5
10


Compressed signal
amplitude
sample
1
0.5
0
-0.5
-1
-10
-5
0
5
10


Time-folded signal
amplitude
sample
1
0.5
0
-0.5
-1
-10
-5
0
5
10

EXPERIMENT NO.3

Aim: Generate the following continuous time domain signal and plot using MATLAB (i) Sine/Cosine signal (ii) Sinc function (iii) Triangle waveform (iv) Rectangular Pulse (v) Sawtooth Waveform (vi) Exponential function

Code:

```
t= -5:0.01:5;


x1 = sin(2*pi*0.5*t);
figure;
subplot(3,2,1);
plot(t, x1);
xlabel("time");
ylabel("amplitude");
title("Sine wave");


x2 = sinc(t);
subplot(3,2,2);
plot(t, x2);
xlabel("time");
ylabel("amplitude");
title("Sinc function");


x3 = tripuls(t, 4);
subplot(3,2,3);
plot(t, x3);
xlabel("time");
ylabel("amplitude");
title("Triangle waveform");

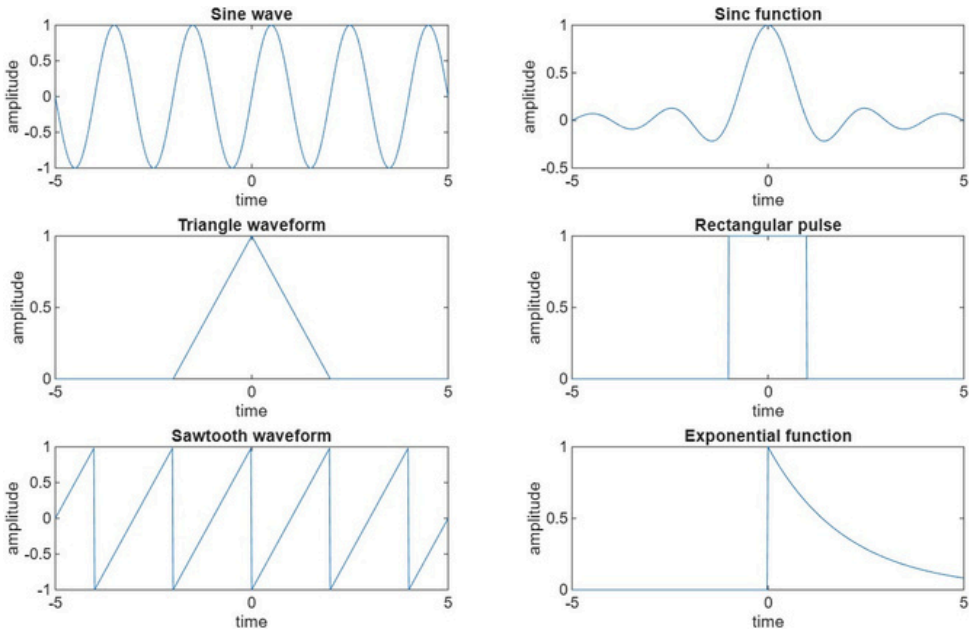

x4 = rectpuls(t, 2);
```

```
subplot(3,2,4);
plot(t, x4);
xlabel("time");
ylabel("amplitude");
title("Rectangular pulse");

x5 = sawtooth(2*pi*0.5*t);
subplot(3,2,5);
plot(t, x5);
xlabel("time");
ylabel("amplitude");
title("Sawtooth waveform");

x6 = exp(-0.5*t) .* (t >= 0);
subplot(3,2,6);
plot(t, x6);
xlabel("time");
ylabel("amplitude");
title("Exponential function");
```

Output



EXPERIMENT NO.4

Aim: To study and verify the following properties of signal/system using MATLAB /Scilab (i) Linearity (ii) Stability (iii) Time invariant. Find the Energy and Power of the given signal. Separate Even and odd components of given signal and plot respectively.

Code:

Linearity:

```
clc;
clear all; close all;
x1 = [1,4,3,6];
x2 = [9,8,7,7];
a = input('Enter a:');
b = input ('Enter b:');
y1 = exp (a*x1);
y2 = exp (b*x2);
y11 = y1 + y2;
y12 = exp (a*x1 + b*x2);
subplot(2,3,1);
stem(x1);
ylabel("Value of X1");
title("Input 1");
grid();
subplot(2,3,2);

stem(x2);
ylabel("Value of X2");
title("Input 2");
grid();
subplot(2,3,3);
stem(y11);
xlabel("Input")
ylabel("Value of y11");
title("Y11");
grid();
subplot(2,3,4);
stem(y12);
xlabel("Input")
ylabel("Value of y12");
title("Y12");
grid();

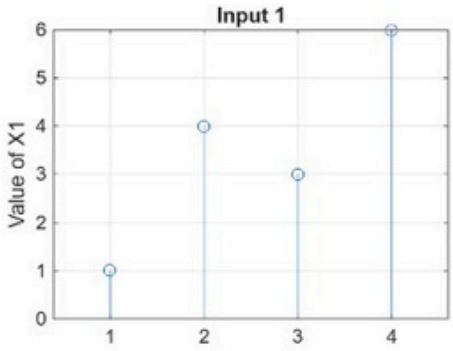
y13 = y11 - y12;
subplot(2,3,5);
stem(y13);
xlabel("Input");
ylabel("Y13");
title("Y13");
if (abs(y11-y12)<= 0.0001)

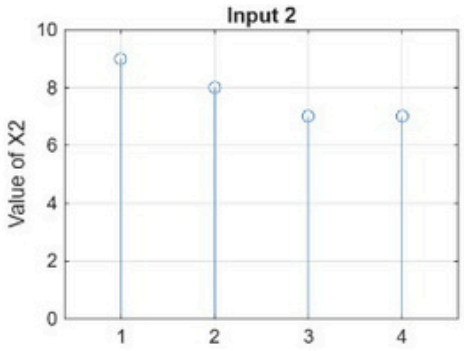
  then
```

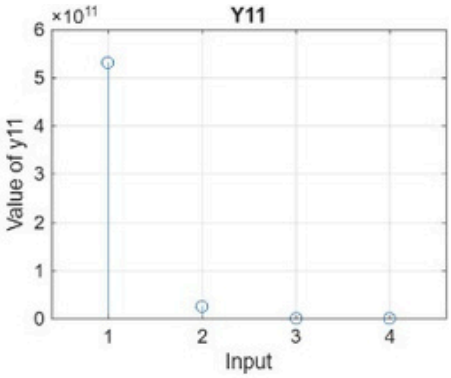
```
disp("System is linear"); else
display("System is non-linear"); end;
```

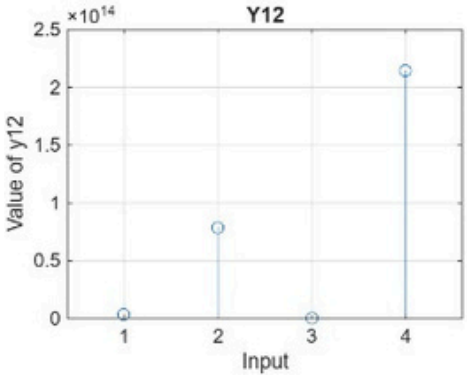
Output:

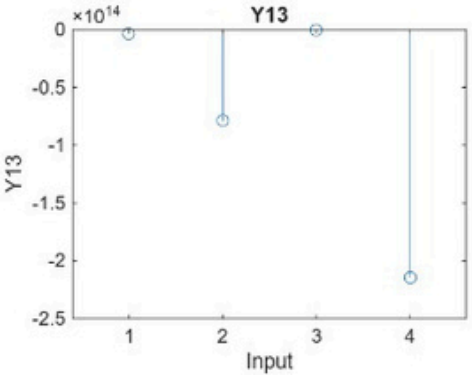
```
Enter a:2
Enter b:3
"System is non-linear"
```











Time variance:

```
clc;

clear all; close all;

x1 = [4, 6, 3, 2, zeros(1,10)];

k = input ("Enter delay:"); x11 = [zeros(1,k), x1];

y11 = exp(x11); % y(n,k);

y12o = exp (x1);

y12 = [ zeros(1,k), y12o];

subplot(2,2,1);

stem(x1);

ylabel("x1");

title("Padded 0 input");

subplot(2,2,2);

stem(x11);

ylabel("x11");

title("delayed input");

subplot(2,2,3);

stem(y11);

xlabel("x11")

ylabel("y11");

title("delayed input and then output");

subplot(2,2,4);

stem(y12);

xlabel("x11")

ylabel("y12");

title("delayed output");

tolerance=0.0001;

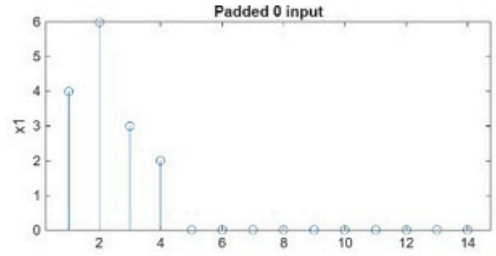
if(abs(y11-y12)<=tolerance)

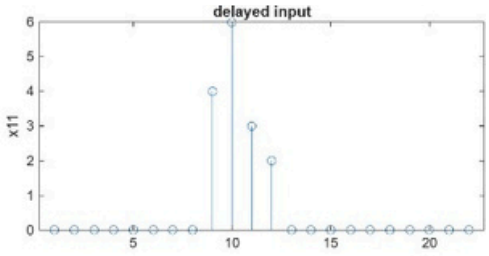
then
```

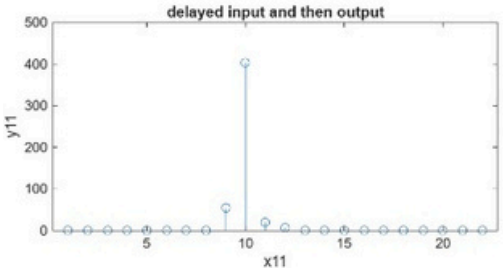
```
disp('the system is time invariant');
else
disp('the system is time variant');
end
```

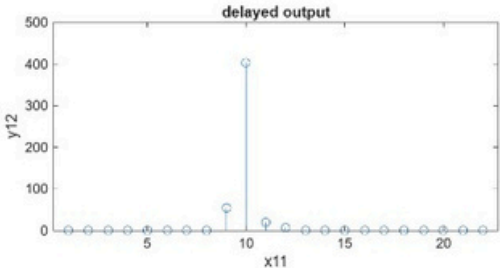
Output:

```
Enter delay: 8
the system is time variant
```









Stability:

```
clc;
clear all;
close all;
x4=[3,-2,4,-2,3];
E=sum(abs(x4).^2);
disp(E);
% stability
```

```
if(E<inf)
   disp('system is stable');
else
   disp('system is unstable');
end
```

Output:

```
 42
system is stable
```

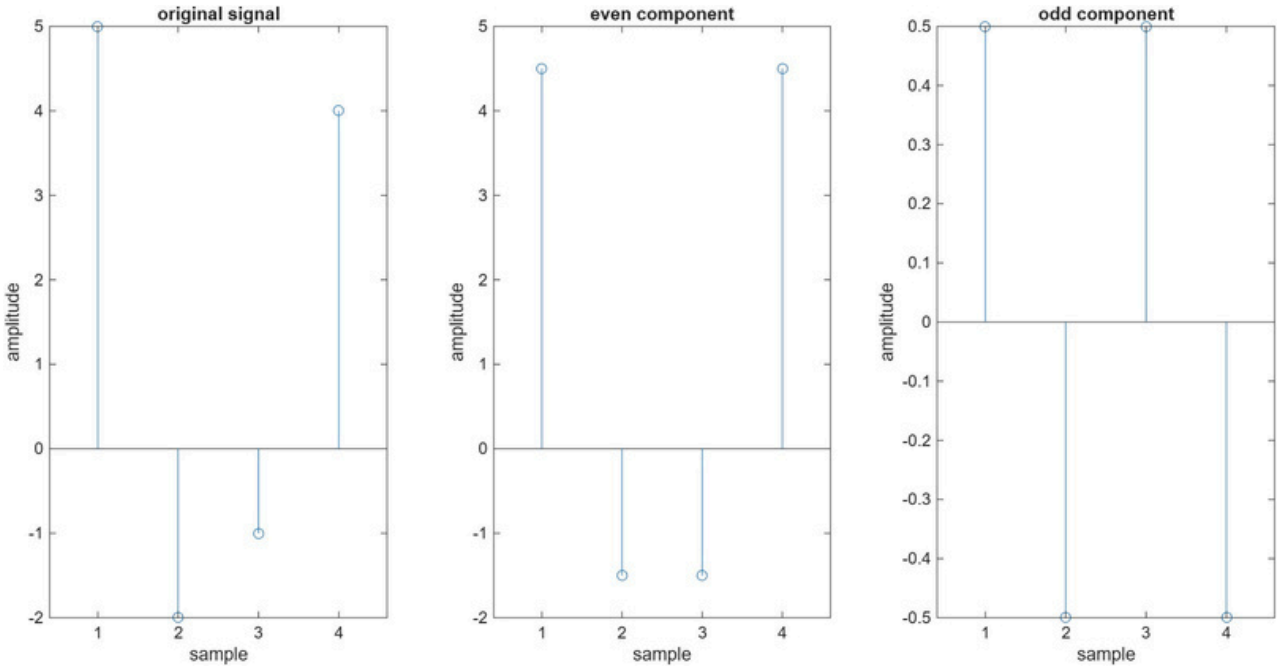
ODD & EVEN:

Code:

```
clc;
clear all;
close all;
X=[5,-2,-1,4];              xe=
(X+X(end:-1:1))/2;     xo=
(X-X(end:-1:1))/2;
subplot(1,3,1);
stem(X);
xlabel("sample");
ylabel("amplitude");
title("original      signal");
subplot(1,3,2);
stem(xe);
xlabel("sample");
ylabel("amplitude");
title("even   component");
subplot(1,3,3);
```

```
stem(xo);
xlabel("sample");
ylabel("amplitude");
title("odd component");
```

Output:



EXPERIMENT NO.5

Aim:To Study convolution of following different sequences using MATLAB/Scilab (without using conv command) and plot the resultant signal. i Convolution between two rectangular signals. ii Convolution between rectangular and Triangular signals. iii Convolution between two Triangular signals.

Code:

Main:

```
clc;
clear all;
close all;


fori=1:3
   fprintf('Case %d:\n', i);
   s1=input("Enter first signal: ");
   s2=input("Enter second signal: ");
   y=Conv_user(s1, s2);


   figure;
   subplot(3,1,1);
   stem(s1);
   xlabel("sample");
   ylabel("amplitude");
   title("signal 1");


   subplot(3,1,2);
   stem(s2);
   xlabel("sample");
   ylabel("amplitude");
   title("signal 2");
```

```
   subplot(3,1,3);
   stem(y);  xlabel("N");
   ylabel("amplitude");
   title("Convolution");


end


function y = Conv_user(x,h)
l1 = length(x);
l2 = length(h);
l3 = l1 + l2 - 1;
x1 = [x, zeros(1, l2)];
h1 = [h, zeros(1, l1)];
y = zeros(1, l3);
for n = 1:l3
   sum = 0;
   for k = 1:n
      sum = sum + x1(k) * h1(n - k + 1);
   end
   y(n) = sum;
end
end
```

Output:

Case 1:

Enter first signal: [1,1,1]

Enter second signal: [2,3,1]

Case 2:

Enter first signal: [2,1,2]

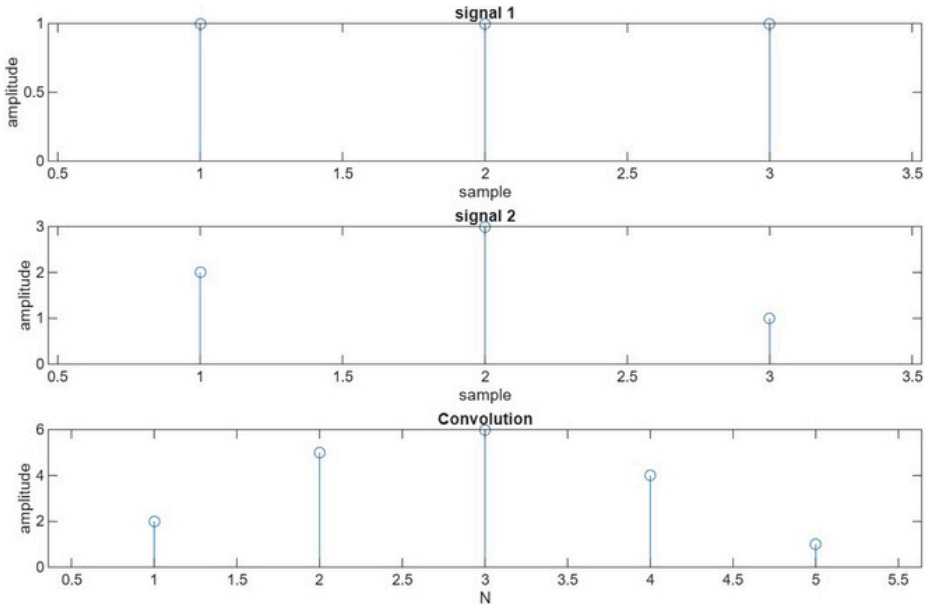
Enter second signal: [1,3,2]

Case 3:

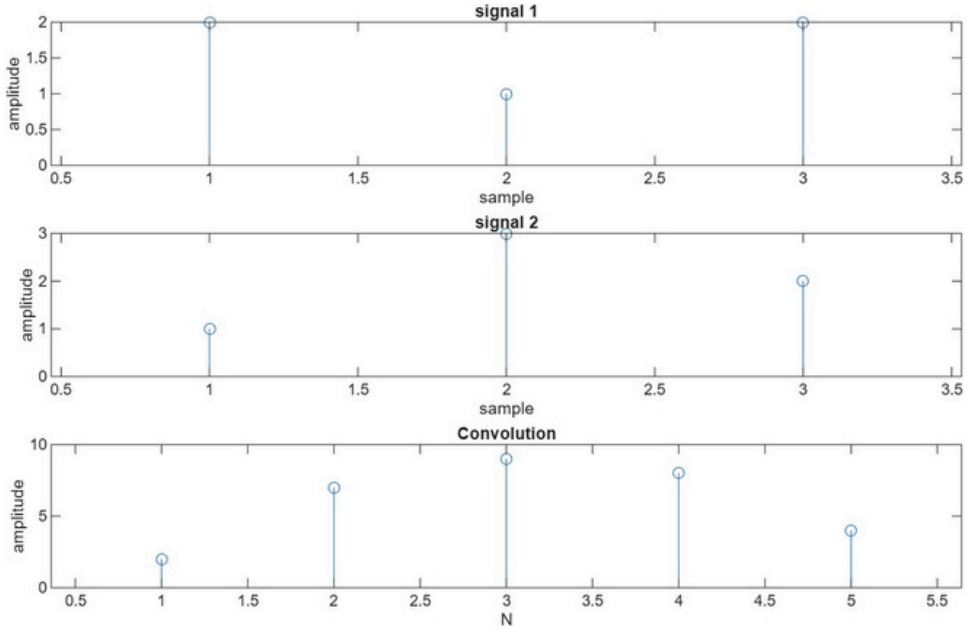
Enter first signal: [1,3,2]

Enter second signal: [1,2,1]

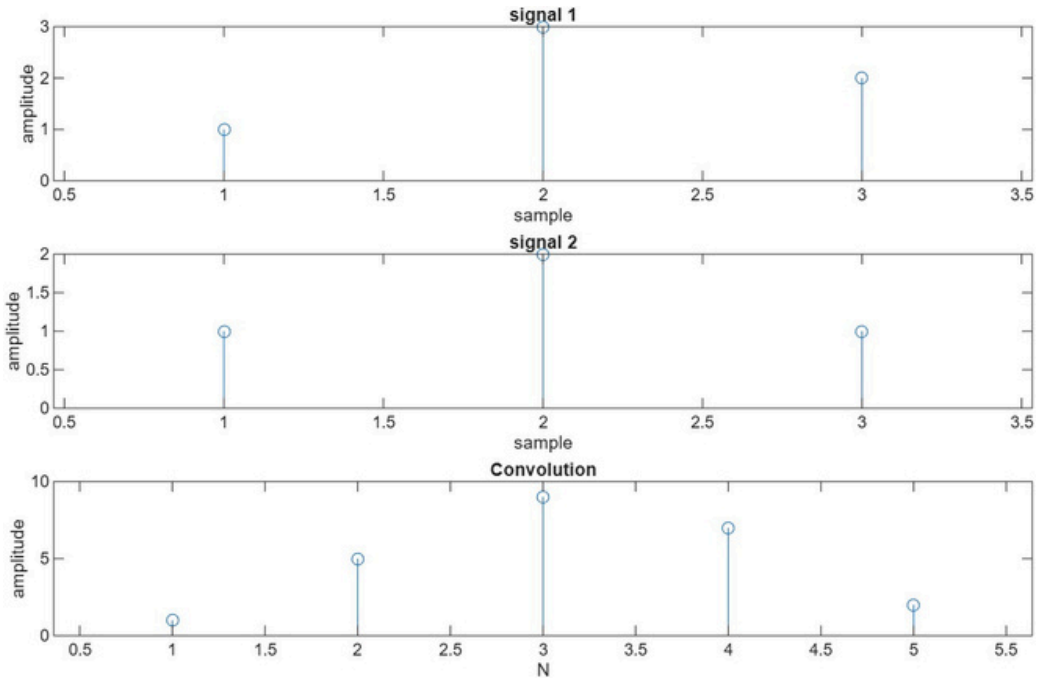
Case 1:



Case 2:



Case 3:


signal 1
amplitude
sample
signal 2
amplitude
sample
Convolution
amplitude
N

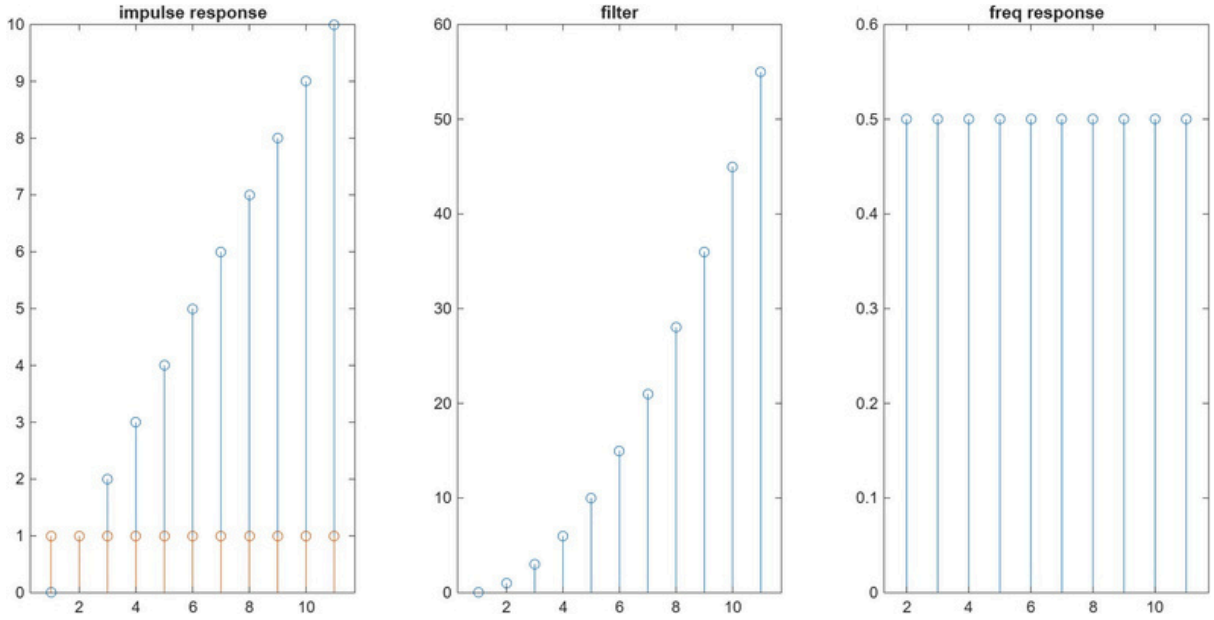
EXPERIMENT NO.6

Aim: To study impz, filter and freqz properties of signal using MATLAB/Scilab.

Code:

```
clc;
clear all;
close all;
b=[1];
a=[1 -1];
n=0:1:10;
subplot(1,3,1);
stem(n);
hold on;
stem(impz(b,a,n));
title("impulse response");
subplot(1,3,2);
stem(n);
stem(filter(b,a,n));
title("filter");
subplot(1,3,3);
stem(n);
stem(freqz(b,a,n));
title("freq response");
```

Output:



EXPERIMENT NO.7

Aim: To Study the Fourier series of Square, Sawtooth and Triangular waves using MATLAB/Scilab.

Code:

Square:

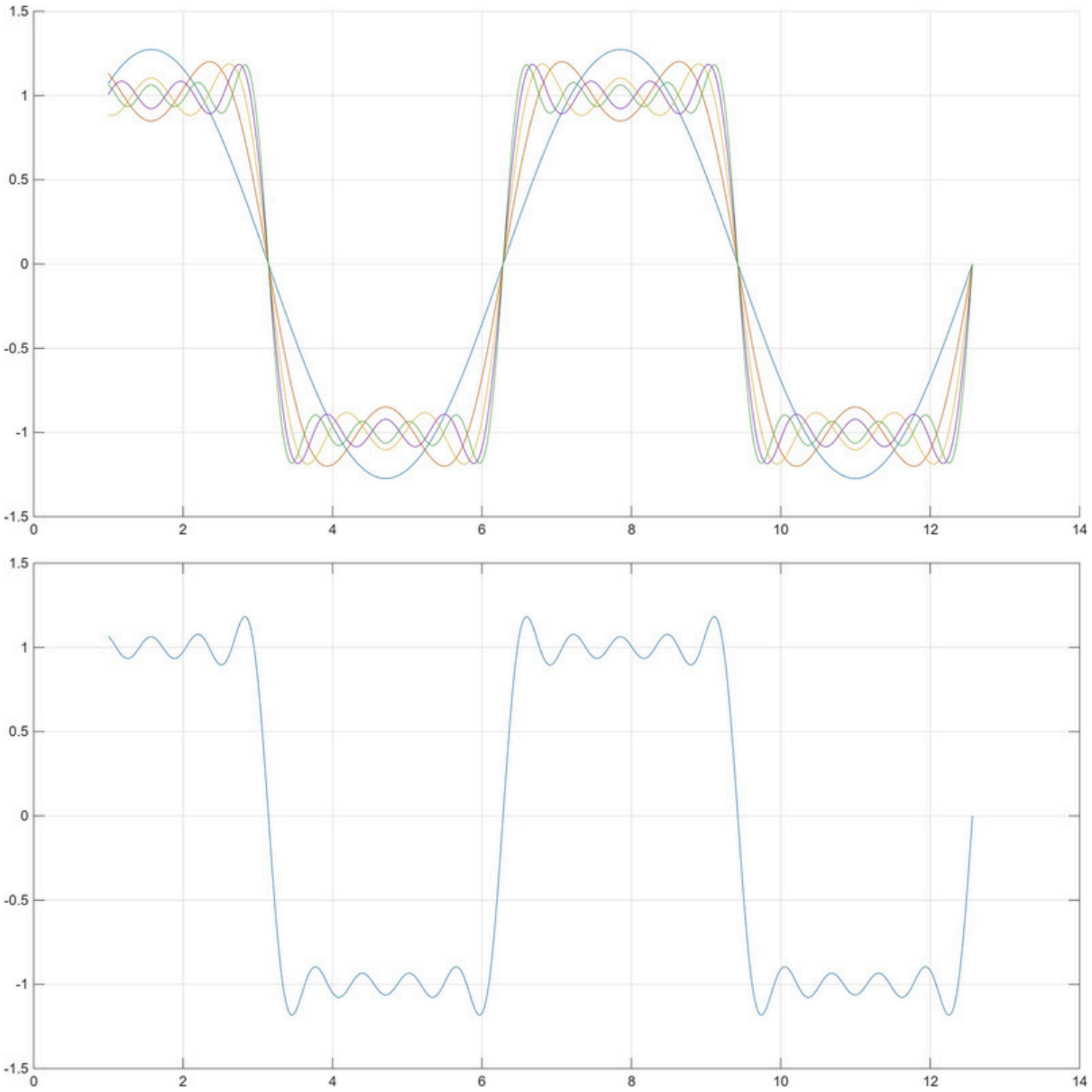
```
clc;
clear  all;
close all;

T = 2*pi;
W0 = (2*pi)/T;
t = linspace(1,4*pi,1000);
N = 10;

sq_wave = zeros(1,length(t));
grid on;
figure(1)
for n = 1:N
   if mod(n,2) == 1
   sq_wave = sq_wave + ((4/pi)*(sin(n*W0*t)/n));
   hold on;
   plot(t,sq_wave);
   end
end

figure(4)
plot(t,sq_wave);
grid on;
```

Output:



Triangular:

Code:

```
clc;
clear all;
close all;

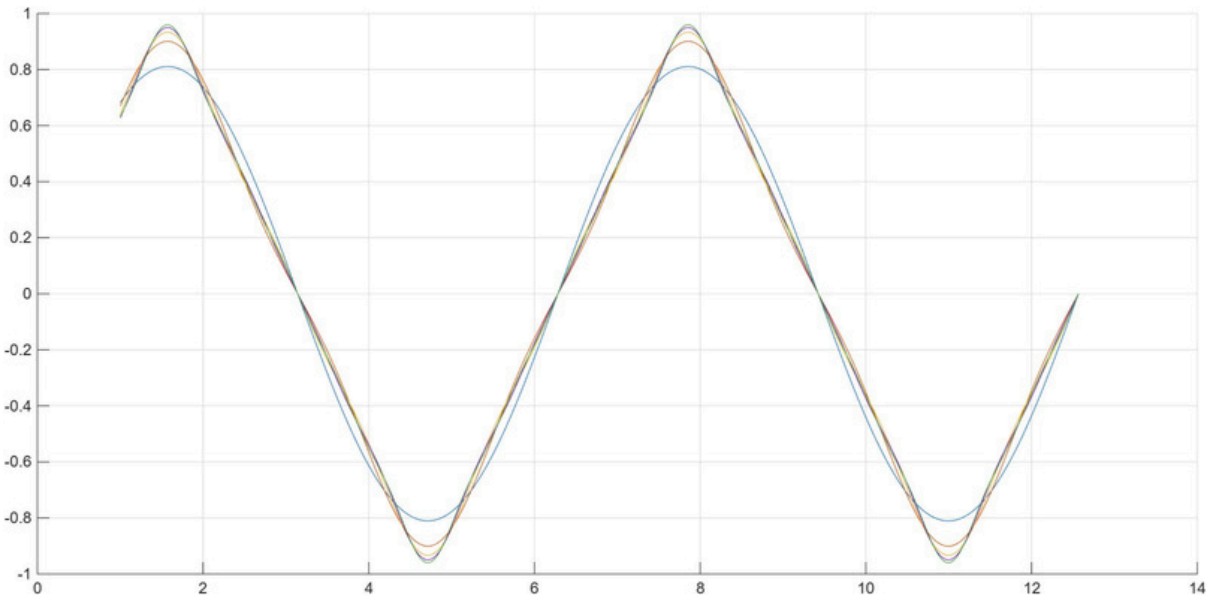

T = 2*pi;
W0 = (2*pi)/T;
```

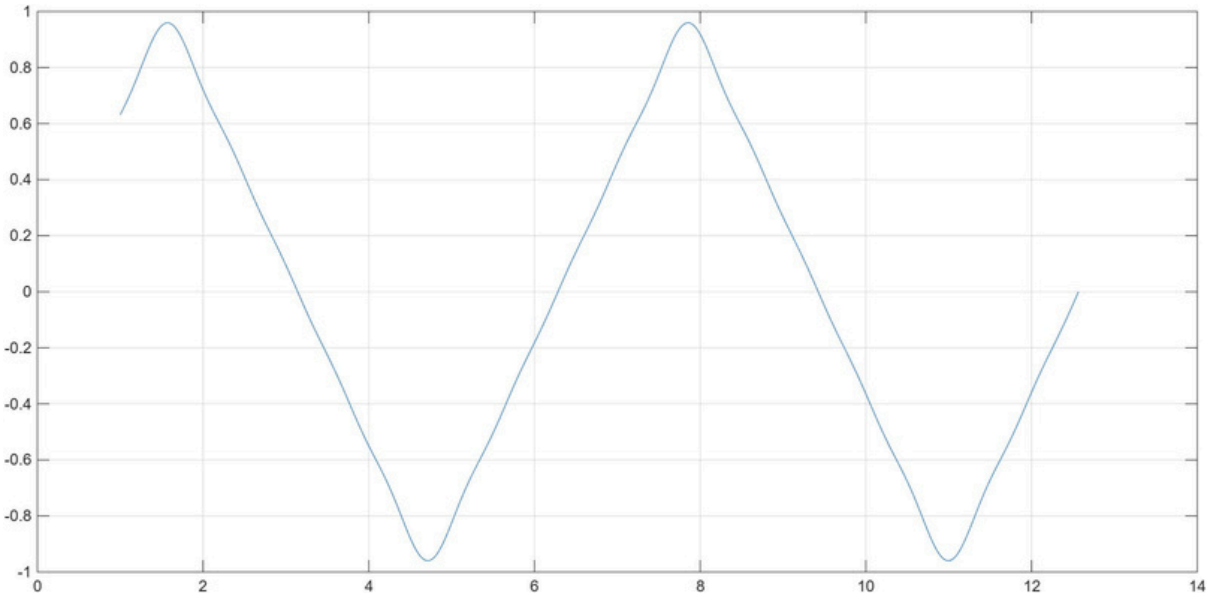
```
t = linspace(1,4*pi,1000);
N = 10;

tg_wave = zeros(1,length(t));


grid on;
figure(1)
for n = 1:N
   if mod(n,2) == 1
      tg_wave = tg_wave + ((8/(pi^2))*(((-1)^((n-1)/2))*(sin(n*W0*t)/(n^2))));
      hold on;
      plot(t,tg_wave);
   end
end

figure(2)
plot(t,tg_wave);
grid on; Output:
```





Sawtooth:

Code:

```
clc;
clear     all;
close all;

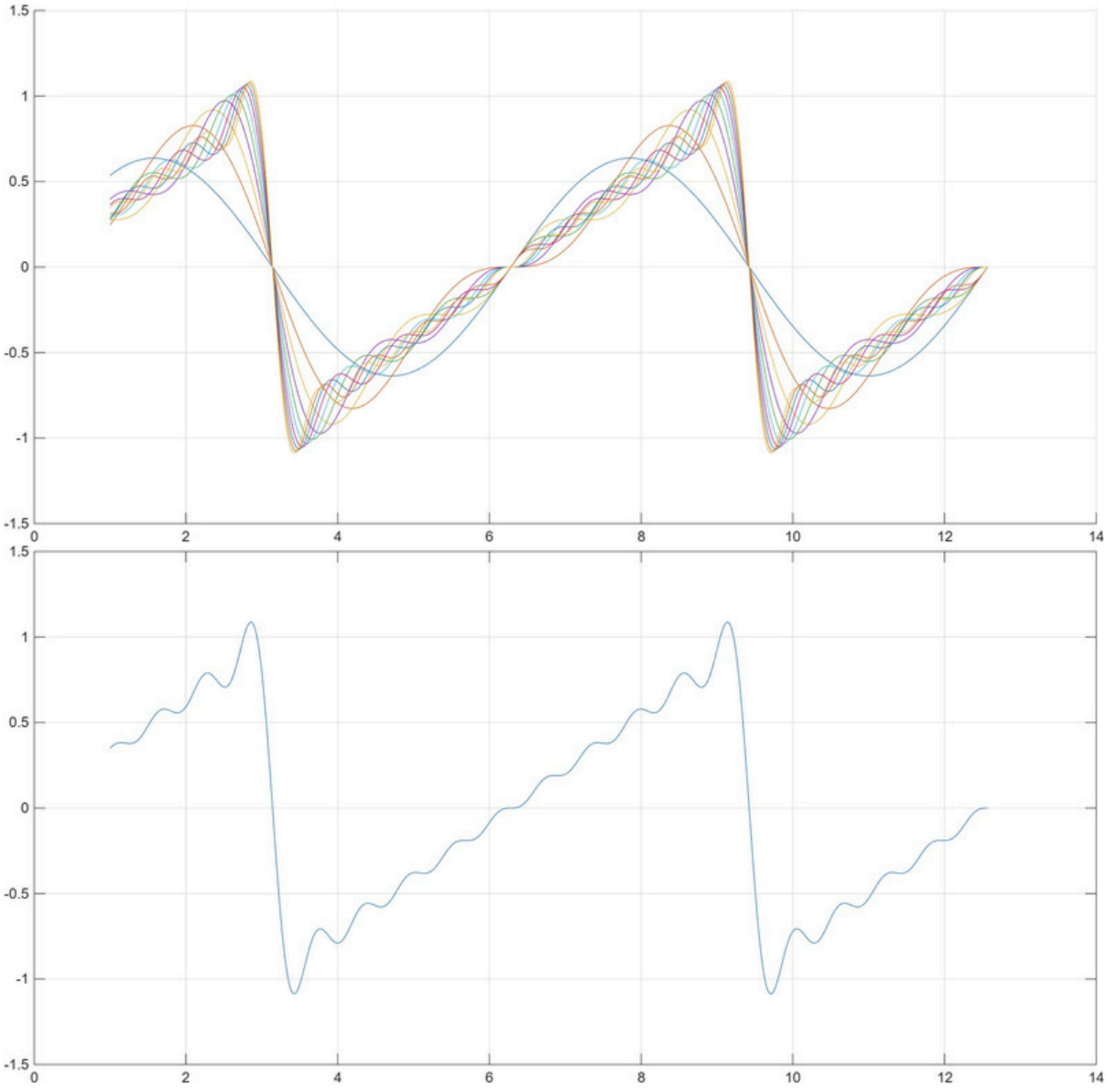

T = 2*pi;
W0 = (2*pi)/T;
t = linspace(1,4*pi,1000);
N = 10;


st_wave = zeros(1,length(t));



grid on;
figure(1)
for n = 1:N
   st_wave = st_wave + ((2/pi)*(((-1)^(n+1))*(sin(n*W0*t)/n)));
   hold on;
```

```
   plot(t,st_wave);
end
figure(2)
plot(t,st_wave);
grid on; Output:
```



EXPERIMENT NO.8:

Aim: To Study Fourier Transform of a signal using FFT and IFFT command and plot magnitude and phase spectra of a signal using MATLA/LabVIEW.

Code:

```
clc;
clear all;
close all;


t=-10:0.01:10;
w=-2*pi:0.01:2*pi;


% 1. Exponential Signal
A = 3;
x_t1 = A.^t;
x_k1 = zeros(size(w));
for k = 1:length(w)
   x_k1(k) = trapz(t, x_t1 .* exp(-1j * w(k) * t));
end
x_kmag1 = abs(x_k1);
x_kangle1 = angle(x_k1);


figure();
subplot(1,3,1);
plot(t, x_t1);
title('Exponential Signal');
xlabel('Time');
ylabel('Amplitude');
grid on;
```

```
subplot(1,3,2);
plot(w, x_kmag1);
title('Magnitude');
xlabel('Frequency');
ylabel('Amplitude');
grid on;

subplot(1,3,3);
plot(w, x_kangle1);
title('Phase');
xlabel('Frequency');
ylabel('Amplitude');
grid on;

% 2. Sine Wave
x_t2 = sin(t);
x_k2 = zeros(size(w));
for k = 1:length(w)
   x_k2(k) = trapz(t, x_t2 .* exp(-1j * w(k) * t));
end
x_kmag2 = abs(x_k2);
x_kangle2 = angle(x_k2);

figure();
subplot(1,3,1);
plot(t, x_t2);
title('Sinewave');
xlabel('Time');
ylabel('Amplitude');
```

```
grid on;

subplot(1,3,2);
plot(w, x_kmag2);
title('Magnitude');
xlabel('Frequency');
ylabel('Amplitude');
grid on;

subplot(1,3,3);
plot(w, x_kangle2);
title('Phase');
xlabel('Frequency');
ylabel('Amplitude');
grid on;

% 3. Parabolic Wave
x_t3 = t.^2;
x_k3 = zeros(size(w));
for k = 1:length(w)
   x_k3(k) = trapz(t, x_t3 .* exp(-1j * w(k) * t));
end
x_kmag3 = abs(x_k3);
x_kangle3 = angle(x_k3);

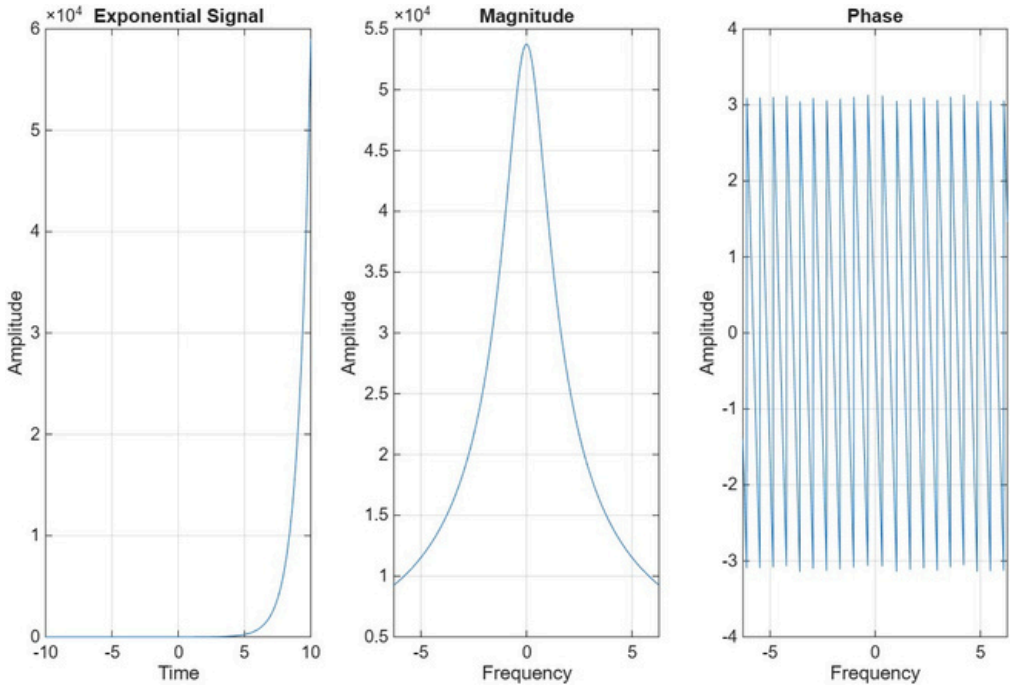
figure();
subplot(1,3,1);
plot(t, x_t3);
title('Parabolic Wave');
```

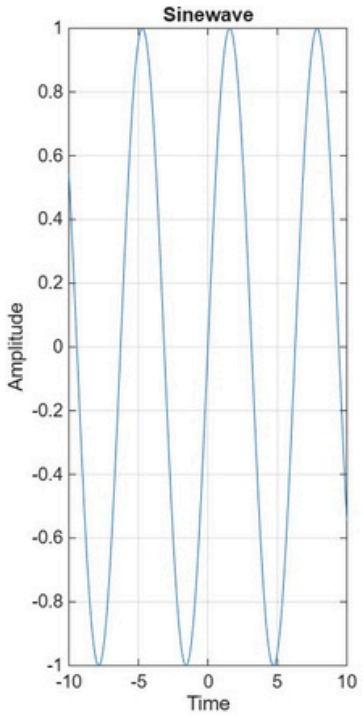
```
xlabel('Time');
ylabel('Amplitude');
grid on;

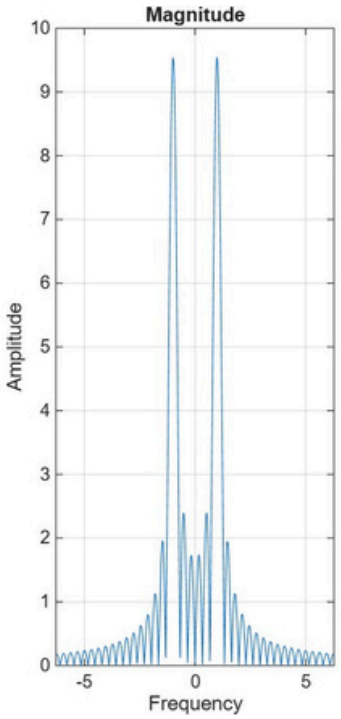
subplot(1,3,2);
plot(w, x_kmag3);
title('Magnitude');
xlabel('Frequency');
ylabel('Amplitude');
grid on;

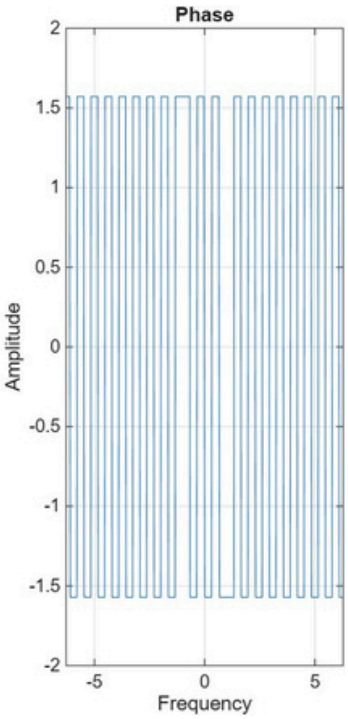
subplot(1,3,3);
plot(w, x_kangle3);
title('Phase');
xlabel('Frequency');
ylabel('Amplitude');
grid on;
```

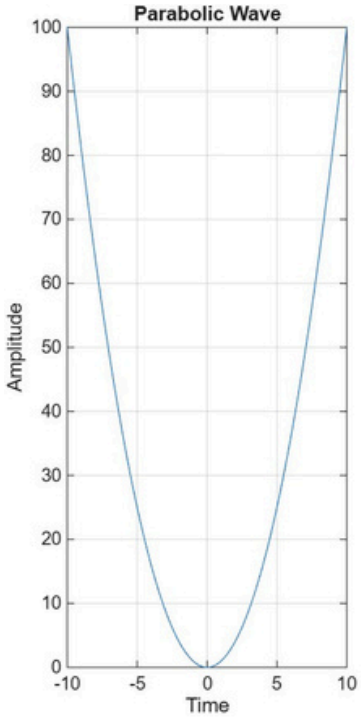
output:

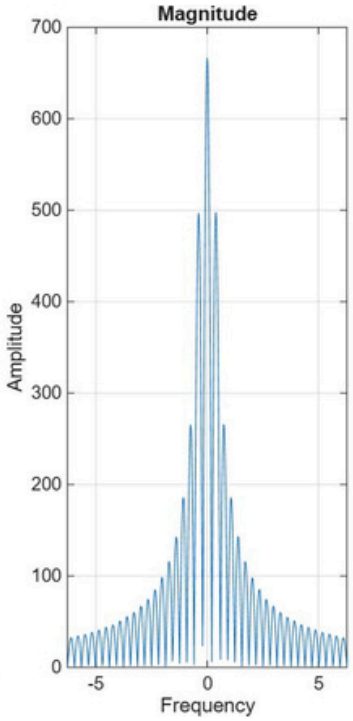


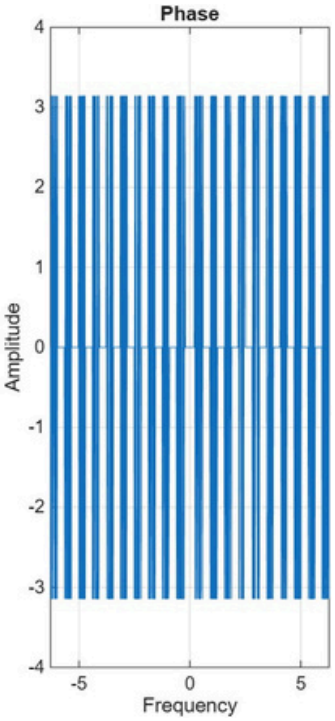

Sinewave
Amplitude
Time


Magnitude
Amplitude
Frequency


Phase
Amplitude
Frequency


Parabolic Wave
Amplitude
Time


Magnitude
Amplitude
Frequency


Phase
Amplitude
Frequency

EXPERIMENT NO.9

Aim: To Study Z Transform and plot region of Convergence using MATLAB Toolbox.

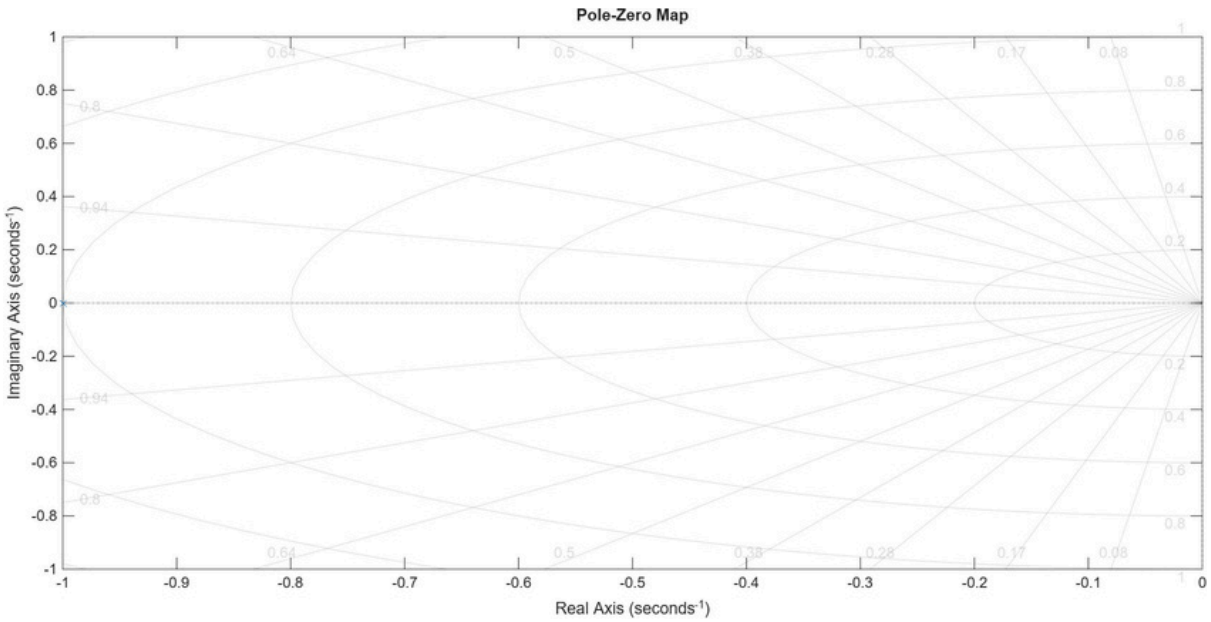
Code:

```
clc;
clear all;
close all;
syms n;
x = sin(n);
fz = ztrans(x);
pretty(fz);
sys = tf([1], [1 2 1]);
pzplot(sys);
grid on;
```

Output:

```
    z sin(1)
------------------
 2
z - cos(1) z 2 + 1
```

Output:



project:Interconnected System Frequency Response

```
clc;
clear;
close all;
% Parameter for all three systems (stable if |a| < 1)
a = 0.8;
% Frequency grid
N = 100;
w = linspace(0, pi, N).';

q = exp(-1j*w); % q = z^-1 on the unit circle

% H1(z): Low-pass filter
b1 = 1 - a;
a1 = [1 -a];

% H2(z): High-pass filter
b2 = [(1 + a)/2, -(1 + a)/2];
a2 = [1 -a];

% H3(z): All-pass filter
b3 = [a 1];
a3 = [1 a];
% Frequency responses
H1 = polyval(fliplr(b1), q) ./ polyval(fliplr(a1), q);
H2 = polyval(fliplr(b2), q) ./ polyval(fliplr(a2), q);
H3 = polyval(fliplr(b3), q) ./ polyval(fliplr(a3), q);

f = w / pi; % Normalized frequency (0 to 1)
% Plot magnitude and phase
figure;
subplot(3,2,1);
plot(f, abs(H1), 'LineWidth', 1.5);
grid on;
title('LPF H_1(z) Magnitude');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H_1|');
```

```
subplot(3,2,2);
plot(f, unwrap(angle(H1)), 'LineWidth', 1.5);
grid on;
title('LPF H_1(z) Phase');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (rad)');

subplot(3,2,3);
plot(f, abs(H2), 'LineWidth', 1.5);
grid on;
title('HPF H_2(z) Magnitude');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H_2|');

subplot(3,2,4);
plot(f, unwrap(angle(H2)), 'LineWidth', 1.5);
grid on;
title('HPF H_2(z) Phase');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (rad)');

subplot(3,2,5);
plot(f, abs(H3), 'LineWidth', 1.5);
grid on;
title('All-pass H_3(z) Magnitude');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H_3|');

subplot(3,2,6);
plot(f, unwrap(angle(H3)), 'LineWidth', 1.5);
grid on;
title('All-pass H_3(z) Phase');
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (rad)');
```

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